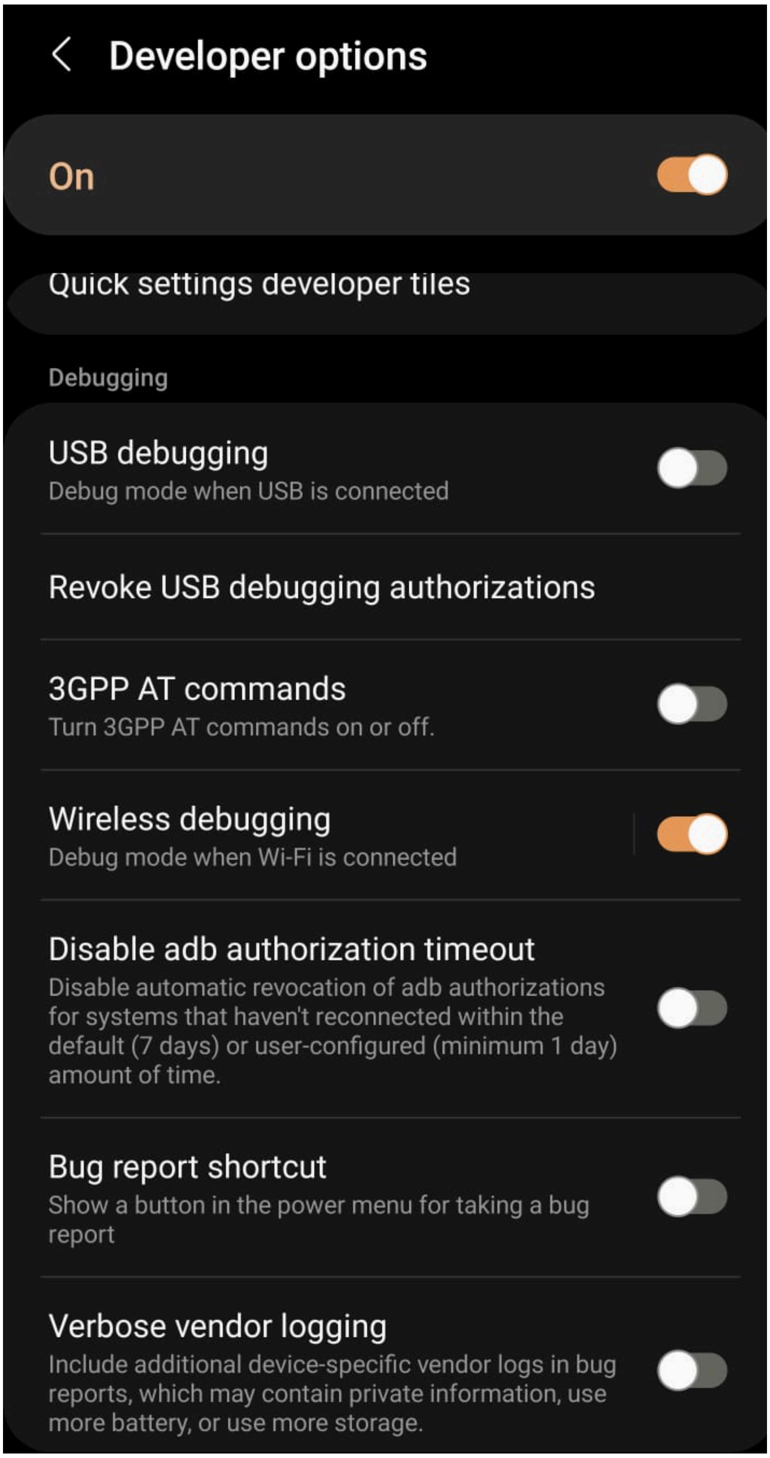
TUTORIAL DEPLOY SERVER DI TERMUX (HP)
MADE BY KRMS FOR GARUDA HACKS 7.0

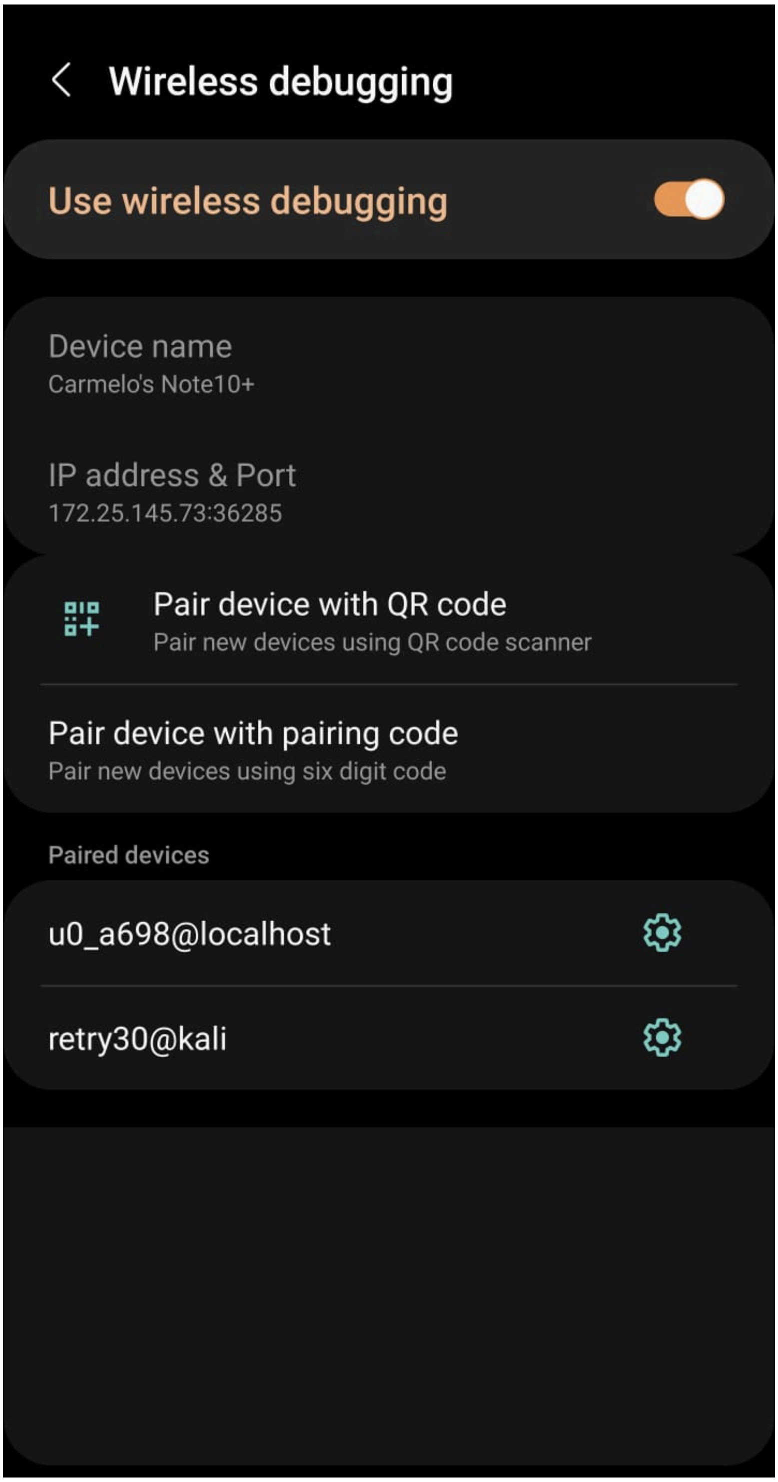
Back end structure

```
GarudaHacks7.0/
│
├── back_end/
│   ├── config/
│   │   └── device.js
│   ├── services/
│   │   ├── adb.js
│   │   ├── location.js
│   │   ├── logger.js
│   │   └── speaker.js
│   ├── node_modules/
│   ├── package.json
│   ├── package-lock.json
│   └── server.js
│
├── AI/
```

1. GO TO SETTING (YOUR PHONE)



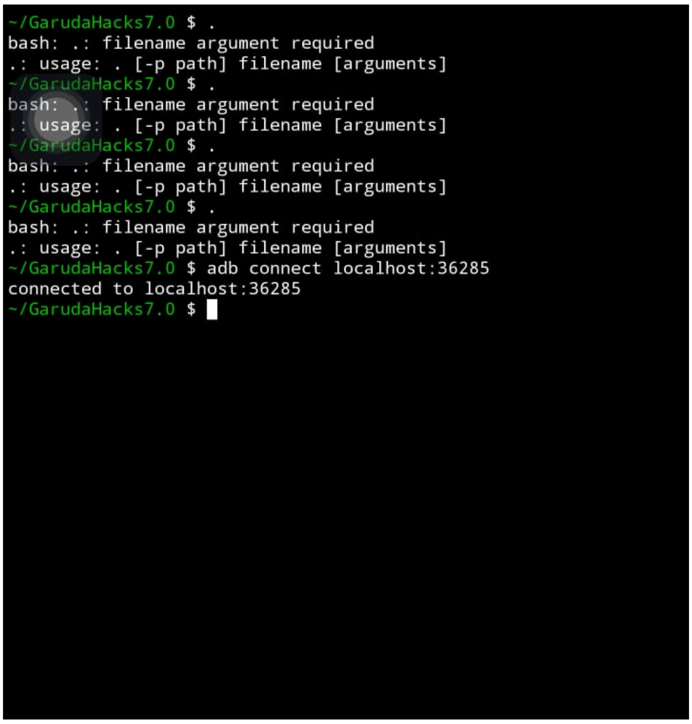
2. FIND WIRELESS DEBUGGING AND CLICK IT!



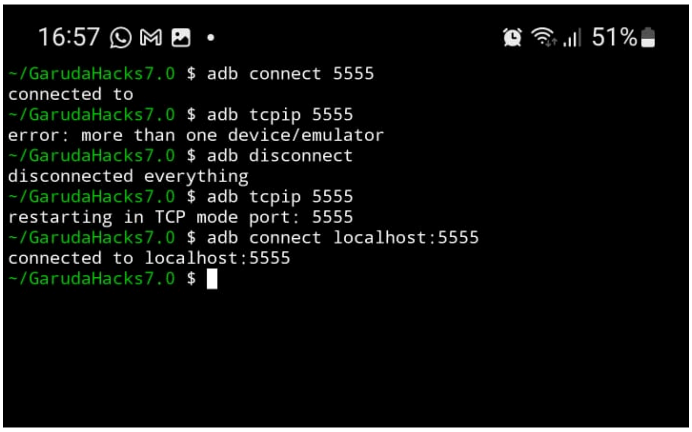
3. Once you in, find 5 number behind double dot symbol for this case, the number is 36285

4. Open your termux

5. Write adb connect localhost:{your number} and PRESS ENTER



6. Write adb disconnect
7. Write adb tcpip 5555
8. adb connect localhost:5555



TUTORIAL TO DEPLOY YOUR SERVER

1. Write node server.js and press enter
   - Remember to follow every step in above

TUTORIAL TO RUN THE THREATS DETECTOR

1. Install Python

- Python 3.10.x is recommended.
- Make sure Python is added to your system PATH.

2. Create a Virtual Environment (Recommended)

terminal: python -m venv .venv

Activate it:
Windows: .venv\Scripts\activate
Linux: source .venv/bin/activate

3. Install Required Libraries

Install all dependencies:

pip install ultralytics
pip install mediapipe
pip install opencv-python
pip install torch torchvision torchaudio (If you have an NVIDIA GPU with CUDA installed, install with **pip install torch torchvision --index-url https://download.pytorch.org/whl/cu128**)
pip install numpy

4. Required Model Files

(Feel free to use the files from the repo :))

pose_landmarker.task # MediaPipe Pose model
yolo11s.pt # YOLO11 COCO model
yolo11gun2.pt # Custom trained pistol model

Project Structure:

```
project/
│
├── main.py
├── detector.py
├── pose.py
├── person.py
├── person_manager.py
├── action.py
├── threat.py
├── recorder.py
├── ui.py
│
├── pose_landmarker.task
├── yolo11s.pt
├── yolo11gun2.pt
│
└── evidence/
```

The **evidence** folder will automatically store screenshots whenever a critical threat is detected.

Running the Program:

Activate the virtual environment first.
run this on terminal: **.venv\Scripts\activate**

After that run: **python main.py**

How the Program Works:

1. The webcam captures live video frames.

2. MediaPipe detects all visible people and estimates their body pose.

3. Every detected person is assigned a unique ID for tracking.

4. The action recognition module classifies each person's activity (standing, walking, running, punching, etc.).

5. YOLO11 detects general objects (e.g., backpack) while the custom YOLO model detects pistols.

6. Detected objects are assigned to the corresponding person based on their bounding box.

7. The threat script computes a threat score using both detected actions and detected objects.

8. When a person's threat score reaches the **CRITICAL** level, the system automatically saves an evidence screenshot.

9. The user interface displays:

- Person ID
- Action
- Threat level
- Detected objects
- FPS
- Alarm indicator (during critical threats)